

# ASSIGNMENT NO.5: Aggregate Functions and Data Analysis

## QUESTION.1

Q1) Create a MySQL database named CollegeDB and a table named StudentResult.

The table should contain the following columns: Column Name Data Type StudentID Constraint INT StudentName VARCHAR(50) Course PRIMARY KEY City VARCHAR(30) VARCHAR(30) Marks FeesPaid INT DECIMAL(10,2) Insert at least 10 records into the table.

### Query

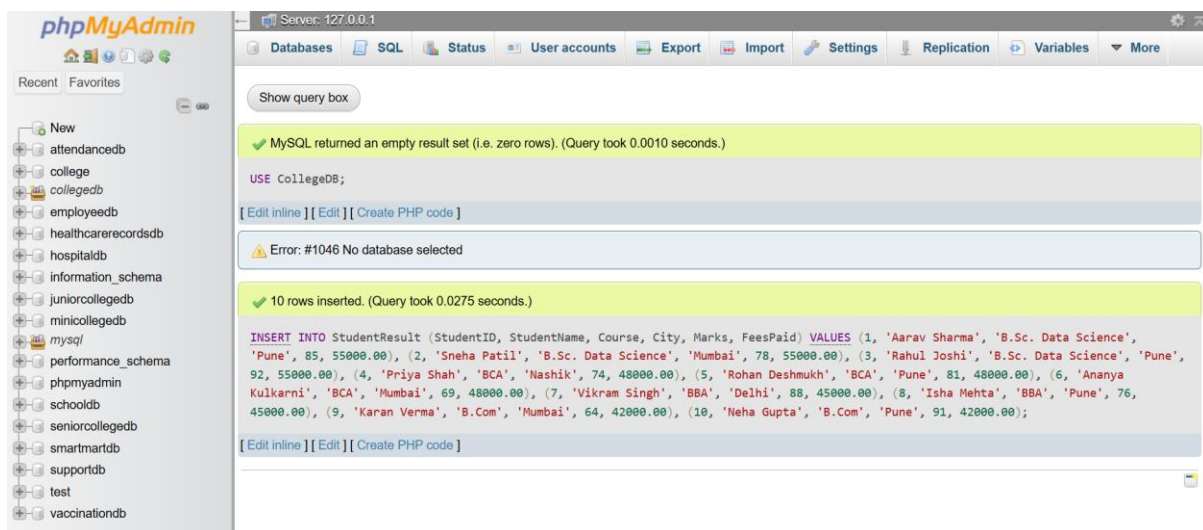
```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE StudentResult (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    Course VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT,  
    FeesPaid DECIMAL(10,2)  
);
```

```
INSERT INTO StudentResult  
(StudentID, StudentName, Course, City, Marks, FeesPaid)  
VALUES  
(1, 'Aarav Sharma', 'B.Sc. Data Science', 'Pune', 85, 55000.00),  
(2, 'Sneha Patil', 'B.Sc. Data Science', 'Mumbai', 78, 55000.00),
```

(3, 'Rahul Joshi', 'B.Sc. Data Science', 'Pune', 92, 55000.00),  
 (4, 'Priya Shah', 'BCA', 'Nashik', 74, 48000.00),  
 (5, 'Rohan Deshmukh', 'BCA', 'Pune', 81, 48000.00),  
 (6, 'Ananya Kulkarni', 'BCA', 'Mumbai', 69, 48000.00),  
 (7, 'Vikram Singh', 'BBA', 'Delhi', 88, 45000.00),  
 (8, 'Isha Mehta', 'BBA', 'Pune', 76, 45000.00),  
 (9, 'Karan Verma', 'B.Com', 'Mumbai', 64, 42000.00),  
 (10, 'Neha Gupta', 'B.Com', 'Pune', 91, 42000.00);



## QUESTION2

Q2) Write MySQL queries to find the following from the StudentResult table:  
 Requirement Total number of students Total fees paid Average marks Lowest marks Highest marks

### QUERY

SELECT COUNT(\*) AS TotalStudents

FROM StudentResult;

SELECT SUM(FeesPaid) AS TotalFeesPaid

FROM StudentResult;

SELECT AVG(Marks) AS AverageMarks

FROM StudentResult;

SELECT MIN(Marks) AS LowestMarks

FROM StudentResult;

SELECT MAX(Marks) AS HighestMarks

FROM StudentResult;

The screenshot shows the phpMyAdmin SQL interface. On the left is a sidebar with a list of databases including 'collegedb', 'employeeedb', 'healthcarerecordsdb', 'hospitaldb', 'information\_schema', 'juniorcollegedb', 'minicollegedb', 'mysql', 'performance\_schema', 'phpmyadmin', 'schooldb', 'seniorcollegedb', 'smartmartdb', 'supportdb', 'test', and 'vaccinationdb'. The main panel displays a message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green bar indicates 'Your SQL query has been executed successfully.' The query shown is: `SELECT COUNT(*) AS TotalStudents FROM StudentResult;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A section titled 'Extra options' shows the result: **TotalStudents** with a value of 10. At the bottom, there is a 'Query results operations' bar with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

The screenshot shows the phpMyAdmin SQL interface. On the left is a sidebar with a list of databases including 'New', 'attendanceedb', 'college', 'collegedb', 'collegedb\_assignment', 'employeeedb', 'healthcarerecordsdb', 'hospitaldb', 'information\_schema', 'juniorcollegedb', 'minicollegedb', 'mysql', 'performance\_schema', 'phpmyadmin', 'schooldb', 'seniorcollegedb', 'smartmartdb', 'supportdb', 'test', and 'vaccinationdb'. The main panel displays a message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green bar indicates 'Showing rows 0 - 0 (1 total, Query took 0.0014 seconds.)'. The query shown is: `SELECT SUM(FeesPaid) AS TotalFeesPaid FROM StudentResult;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A section titled 'Extra options' shows the result: **TotalFeesPaid** with a value of 483000.00. At the bottom, there is a 'Query results operations' bar with buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

phpMyAdmin

Recent Favorites

New

- attendancedb
- college
- collegedb
- collegedb
- collegedb\_assignment
- employeedb
- healthcarerecordsdb
- hospitaldb
- information\_schema
- juniorcollegedb
- minicollegedb
- mysql

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings

Showing rows 0 - 0 (1 total, Query took 0.0007 seconds.)

`SELECT AVG(Marks) AS AverageMarks FROM StudentResult;`

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

**AverageMarks**

79.8000

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

phpMyAdmin

Recent Favorites

New

- attendancedb
- college
- collegedb
- collegedb
- collegedb\_assignment
- employeedb
- healthcarerecordsdb
- hospitaldb
- information\_schema
- juniorcollegedb
- minicollegedb
- mysql
- performance\_schema
- phpmyadmin

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings

Showing rows 0 - 0 (1 total, Query took 0.0014 seconds.)

`SELECT MIN(Marks) AS LowestMarks FROM StudentResult;`

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

**LowestMarks**

64

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

phpMyAdmin

Recent Favorites

New

- attendancedb
- college
- collegedb
- collegedb
- collegedb\_assignment
- employeedb
- healthcarerecordsdb
- hospitaldb
- information\_schema
- juniorcollegedb
- minicollegedb
- mysql
- performance\_schema
- phpmyadmin
- schooldb
- seniorcollegedb

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are r

Showing rows 0 - 0 (1 total, Query took 0.0036 seconds.)

`SELECT MAX(Marks) AS HighestMarks FROM StudentResult;`

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

**HighestMarks**

92

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

### QUESTION 3

Q3) Write a MySQL query to count the number of students in each course. The output should contain: Column Course TotalStudents

#### QUERY

```
SELECT COURSE, COUNT(*) AS TotalStudents
```

```
FROM studentresult
```

```
GROUP BY course;
```

The screenshot shows the phpMyAdmin interface with the 'SQL' tab selected. The query entered is: `SELECT COURSE, COUNT(*) AS TotalStudents FROM studentresult GROUP BY course;`. The results are displayed in a table with two columns: 'COURSE' and 'TotalStudents'. The results are as follows:

| COURSE             | TotalStudents |
|--------------------|---------------|
| B.Com              | 2             |
| B.Sc. Data Science | 3             |
| BBA                | 2             |
| BCA                | 3             |

### QUESTION 4

Q4) Write a MySQL query to find the average marks of students course-wise. The output should contain: Column Course AverageMarks

#### QUERY

```
SELECT course, AVG(Marks) AS AverageMarks
```

```
FROM StudentResult
```

```
GROUP BY course;
```

localhost/phpmyadmin/index.php?route=/server/sql

Server: 127.0.0.1

Databases SQL Status User accounts Export Import Settings Rep

Showing rows 0 - 3 (4 total, Query took 0.0008 seconds.)

```
SELECT course, AVG(Marks) AS AverageMarks FROM StudentResult GROUP BY course;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

|                                           | course             | AverageMarks |
|-------------------------------------------|--------------------|--------------|
| <input type="checkbox"/> Edit Copy Delete | B.Com              | 77.5000      |
| <input type="checkbox"/> Edit Copy Delete | B.Sc. Data Science | 85.0000      |
| <input type="checkbox"/> Edit Copy Delete | BBA                | 82.0000      |
| <input type="checkbox"/> Edit Copy Delete | BCA                | 74.6667      |

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Console

## QUESTION 5

Q5) Write a MySQL query to display only those courses where the number of students is greater than 2. Use: Clause GROUP BY HAVING

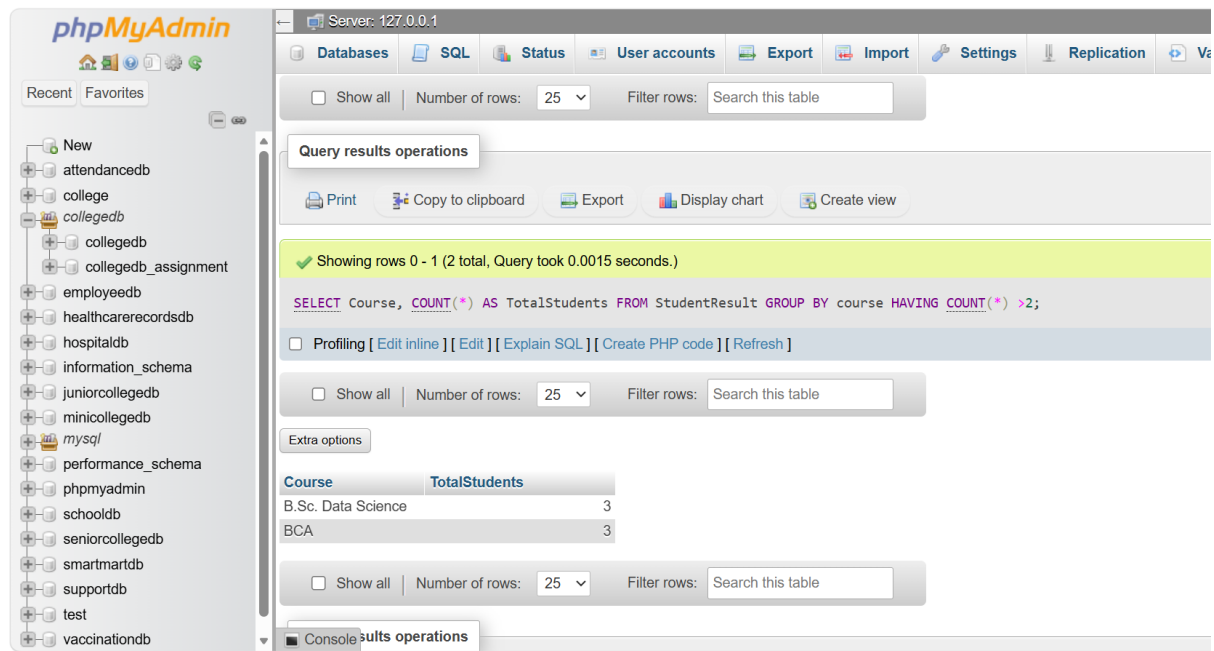
### QUERY

SELECT Course, COUNT(\*) AS TotalStudents

FROM StudentResult

GROUP BY course

HAVING COUNT(\*) >2;



## QUESTION6

Q6) Write a MySQL query to analyze course performance. The output should display: Column Course Column TotalStudents AverageMarks LowestMarks HighestMarks

## QUERY

SELECT

Course,

COUNT(\*) AS TotalStudents,

ROUND(AVG(Marks), 2) AS AverageMarks,

MIN(Marks) AS LowestMarks,

MAX(Marks) AS HighestMarks

FROM StudentResult

GROUP BY Course

ORDER BY AverageMarks DESC;

phpMyAdmin

Recent Favorites

New

attendanceodb

college

collegedb

collegedb\_assignment

employeedb

healthcarerecordsdb

hospitaldb

information\_schema

juniorcollegedb

minicollegedb

mysql

performance\_schema

phpmyadmin

schooldb

seniorcollegedb

smartmartdb

supportodb

test

vaccinationdb

Server: 127.0.0.1

DatabasesSQLStatusUser accountsExportImportSettingsReplicationVariablesMore

Showing rows 0 - 3 (4 total, Query took 0.0020 seconds.)

SELECT Course, COUNT(\*) AS TotalStudents, ROUND(AVG(Marks), 2) AS AverageMarks, MIN(Marks) AS LowestMarks, MAX(Marks) AS HighestMarks

FROM StudentResult GROUP BY Course ORDER BY AverageMarks DESC;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

| Course             | TotalStudents | AverageMarks | LowestMarks | HighestMarks |
|--------------------|---------------|--------------|-------------|--------------|
| B.Sc. Data Science | 3             | 85.00        | 78          | 92           |
| BBA                | 2             | 82.00        | 76          | 88           |
| B.Com              | 2             | 77.50        | 64          | 91           |
| BCA                | 3             | 74.67        | 69          | 81           |

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

PrintCopy to clipboardExportDisplay chartCreate view

Console